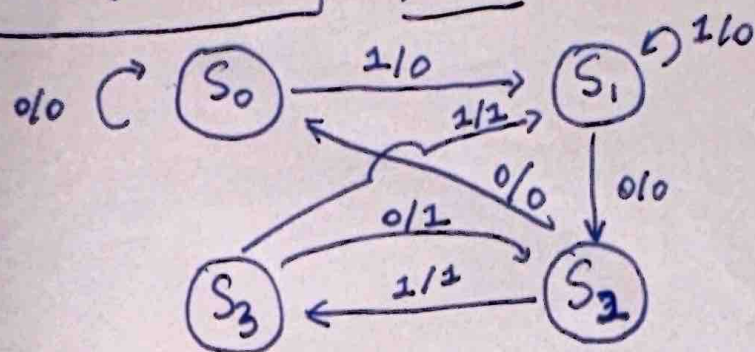


Challenge - 2-A.

MEALY:-



S_0 :- IDLE STATE
(X)

S_1 :- Detects first 1
(X1)

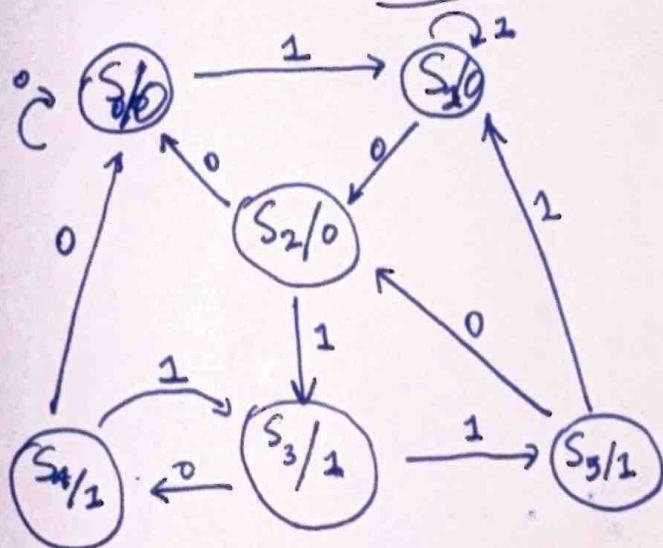
S_2 :- Detects 0 after 1
(X10)

S_3 :- Detects 101 in
last 3 digits
(X101)

After S_3 , for next input

Output is always 1 since
"101" is still detected last
4 digits.

MOORE:-



S_0 :- IDLE STATE
(X)

S_1 :- Detects first 1
(X1)

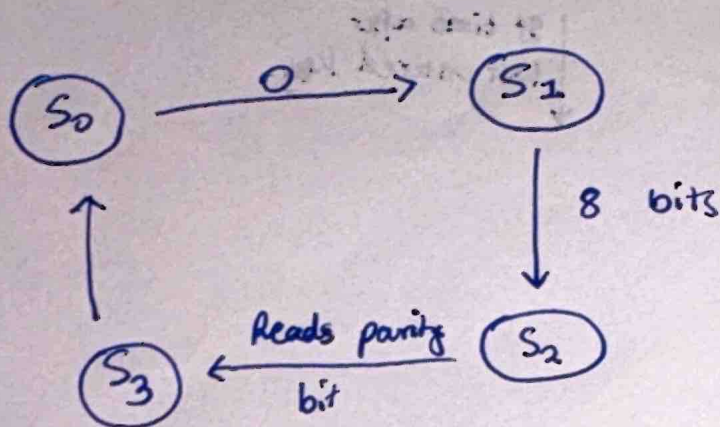
S_2 :- Detects first 0 after 1
(X10)

S_3 :- Detects 101 in last
3 digits
(X101)

S_4 :- Detects 1010 in last
4 digits.

S_5 :- Detects 1011 in last
4 digits.

Challenge -2B



S_0 : IDLE state
(start bit is 1).

S_1 : After start bit is detected, starts to read 8 bits.

S_2 : After reading data bits, reads parity bit.

S_3 : Calculate error in parity, Reads Stop bit.

If any error is found, data outputted is 0. & error bit is high for 1 clock cycle.

